XRPL Validator — Daily Engineering Report · 2026-11-13

Period: 2026-11-13 00:00 to 24:00 US Eastern (2026-11-13 05:00 to 2026-11-14 05:00 UTC); ledgers #108,296,677 to #108,317,996. Every count from the ledger and the validator's logs stops at #108,317,996, closed 04:59:52 UTC. Generated 2026-11-14 00:21 EST by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded — coverage outside planned maintenance 99.109% (objective 99.9%): 190 ledgers unsigned.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,130 of the 21,320 ledgers the network closed (99.109%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 190 outside maintenance (4 incidents).
- **Native engine:** 1 engine receipt: #108,298,587 — open.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 14 slow-close periods; the longest, 15:27 UTC–15:29 UTC, closed ledgers up to 68 s apart.
- **Independence program:** 16 of 30 qualifying days (snapshot at 05:21:14 UTC).

Service levels

Measure	Value
Ledgers closed by the network	21,320
Validations issued	21,130
Coverage (validations / ledgers closed)	99.109%
Validations matching the network's validated ledger hash	21,130 of 21,130
Unsigned during planned restarts	0
Unsigned outside maintenance	190
Coverage outside planned maintenance (objective 99.9%)	99.109%
Restarts	0
Basis: our own ledger hash	21,107
Basis: our own hash, witnessed by the reference node (quorum short)	23
External agreement, data.xrpl.org (UTC day 2026-11-13)	0.99002 (204 missed of 20,451)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,296,677–#108,317,996), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	1
Step 4 · peer data	0
Step 4 · validations	106
Step 5 · own hash	0
Writer	1

Live counters, snapshot at 05:21:14 UTC (since the last restart, 2026-11-05 00:49 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	192,800 ledgers, 39,840,175 transactions	192,798 fully matched; results 4, bytes 64, keys 1/12
2 · transaction records	197,138 ledgers, 40,714,369 records	40,714,109 exact; differ 256, tree 3
3 · transaction order	197,350 ledgers (mode: on)	197,344 exact; order 6, results 2, bytes 0, refused 6
4 · ledger data from peers	197,357 ledgers fetched	0 matched; differ 0, failed 1
4 · trusted validations	197,293 ledgers judged	197,154 agreed; differ 0, short of quorum 139
5 · our own ledger hash	197,354 ledgers	197,129 own; differ 3, partial 222
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	07:07 UTC	#108,298,556–#108,298,638 (83)	Step 4 · validations	0–25 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (none signed); see INC-01
R-02	07:09 UTC	#108,298,587	Step 3 · order, Writer	our transaction order differed at position 0; 2 result(s) differed	open; see INC-01
R-03	08:23 UTC	#108,299,680–#108,299,698 (19)	Step 4 · validations	0 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-04	15:27 UTC	#108,305,983–#108,305,984 (2)	Step 4 · validations	25–26 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-05	01:37 UTC	#108,315,024–#108,315,025 (2)	Step 4 · validations	23–27 of the 28 trusted validations needed reached the	explained: network delivery, not an engine defect; the network validated these ledgers

ID	Time	Ledger(s)	Step	Description	Disposition
				validator in time	normally (all signed)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-11-13-01, INC-01 is INC-2026-11-13-01.

Incidents

INC-01 · 109 ledgers unsigned

Window	07:06:32 UTC to 07:13:52 UTC
Ledgers	#108,298,538–#108,298,646 (109)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-01, R-02
Cause	Cause not recorded.

INC-02 · 43 ledgers unsigned

Window	07:14:20 UTC to 07:17:10 UTC
Ledgers	#108,298,652–#108,298,694 (43)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Cause	Cause not recorded.

INC-03 · 16 ledgers unsigned

Window	08:21:41 UTC to 08:22:41 UTC
Ledgers	#108,299,660–#108,299,675 (16)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Cause	Cause not recorded.

INC-04 · 22 ledgers unsigned

Window	08:24:31 UTC to 08:26:00 UTC
Ledgers	#108,299,702–#108,299,723 (22)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Cause	Cause not recorded.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **16 of 30** (snapshot at 05:21:14 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-11-13 — tracked #108,292,246–#108,313,591	Value
Validations	21,156 (21,156 own, 0 borrowed)
Coverage of the tracked ledgers	99.11%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,320 (#108,296,677–#108,317,996)
Mean close interval	4.05 s
Longest close interval	68 s (before #108,305,984)
Closes at least 15 s apart	18
Empty ledgers	0

- Slow closes 08:38:41 UTC–08:39:00 UTC: ledgers #108,299,911, 1 close at least 15 s apart, up to 19 s.
- Slow closes 09:03:32 UTC–09:03:50 UTC: ledgers #108,300,276, 1 close at least 15 s apart, up to 18 s.
- Slow closes 10:34:41 UTC–10:35:40 UTC: ledgers #108,301,650–#108,301,654 (5), 3 closes at least 15 s apart, up to 19 s.
- Slow closes 13:30:20 UTC–13:30:40 UTC: ledgers #108,304,286, 1 close at least 15 s apart, up to 20 s.
- Slow closes 14:04:40 UTC–14:05:00 UTC: ledgers #108,304,774, 1 close at least 15 s apart, up to 20 s.
- Slow closes 14:10:32 UTC–14:10:50 UTC: ledgers #108,304,858, 1 close at least 15 s apart, up to 18 s.
- Slow closes 15:09:11 UTC–15:09:30 UTC: ledgers #108,305,718, 1 close at least 15 s apart, up to 19 s.
- Slow closes 15:27:32 UTC–15:29:00 UTC: ledgers #108,305,984–#108,305,987 (4), 2 closes at least 15 s apart, up to 68 s.

- Slow closes 15:29:03 UTC–15:29:20 UTC: ledgers #108,305,991, 1 close at least 15 s apart, up to 17 s.
- Slow closes 17:34:21 UTC–17:34:40 UTC: ledgers #108,307,824, 1 close at least 15 s apart, up to 19 s.
- Slow closes 17:34:44 UTC–17:35:00 UTC: ledgers #108,307,829, 1 close at least 15 s apart, up to 16 s.
- Slow closes 00:40:21 UTC–00:40:40 UTC: ledgers #108,314,186, 1 close at least 15 s apart, up to 19 s.
- Slow closes 01:37:02 UTC–01:38:20 UTC: ledgers #108,315,025–#108,315,028 (4), 2 closes at least 15 s apart, up to 58 s.
- Slow closes 02:20:31 UTC–02:20:50 UTC: ledgers #108,315,656, 1 close at least 15 s apart, up to 19 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 98.

Open items

- INC-01 (#108,298,538–#108,298,646 (109)): cause not recorded.
- INC-02 (#108,298,652–#108,298,694 (43)): cause not recorded.
- INC-03 (#108,299,660–#108,299,675 (16)): cause not recorded.
- INC-04 (#108,299,702–#108,299,723 (22)): cause not recorded.

Corrections to earlier reports

None.